Grippy: Operational Strategy Report

CVO Final Project - Operations Management (2026F-A02)

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1. Venture Overview & Value Proposition

1.1 Industry Opportunity

Electricians typically operate in dynamic work environments. While some trade contractors have a fixed working environment, electricians are constantly on the move - regularly shifting between job sites while managing an extensive kit of tools and adapting to work in confined spaces. This creates operational inefficiencies driven by the need to transport, organize, locate and access equipment throughout the working day.

While the conventional, upright backpacks address the problem of portability and storage, these force technicians to dig through layered compartments to retrieve the tools buried at the bottom. Most incumbent bags are built for storage and durability. The closest product, Klein Tools' Tool Station Backpack, adds a front flap that folds down into a workstation and padded, adjustable straps[1], but comfort is treated as padding on a storage bag rather than as the design goal. While the current market solutions improve accessibility, a significant gap remains for a solution that brings together optimal visibility of tools with workspace functionality and ergonomic comfort. Back support has been a recurring pain point in our interviews with electricians[2], so incumbents under-serve independent electricians who carry a full kit between job sites every day.

1.2 Proposed Customer and Value Proposition

The proposed solution, the Grippy backpack, leads with an ergonomic support harness that ensures maximum comfort and minimizes physical strain while transiting between sites. The bag transforms from a conventional style into a workspace that opens flat, and its detachable compartments (a base set plus add-on compartments the customer picks at order) can be fixed onto a utility belt for access on the move. Its target users are independent electrical contractors who operate across commercial and residential job sites without a fixed workplace infrastructure. Together, these features maximize tool visibility and reduce operational delays caused by rummaging.

1.3 Uncommon Belief and Value Theory

The common belief in the trade is that a tool bag must trade off capacity, access and carrying comfort. Grippy's uncommon belief is that independent electricians will pay for carrying comfort

[1]Klein Tools, Tradesman Pro Tool Station Backpack (Cat. No. 55482), kleintools.com.
[2]Grippy team interviews with electricians, fall 2026.

and on-the-move access, which incumbents treat as secondary. Following the Value Lab approach[3]. The problem Grippy must solve is to deliver that comfort and access without giving up the capacity and durability incumbents already offer. Its value theory, tested in Section 6.3, is: independent electricians who move between job sites will prefer, at a price close to the Klein Tool Station, a backpack that opens flat, carries detachable compartments for access on the move and reduces carrying strain through an ergonomic support harness. If the theory holds, this segment switches first, followed by other mobile trades.

1.4 Scope and Strategic Choice

The venture would initially focus on a standardized model of the backpack (one body in one size, with a base set of detachable compartments) to minimize production overhead and manufacturing complexity. A contract backpack manufacturer will cut and sew the bag bodies, straps and compartments to Grippy's specification; Grippy will inspect incoming lots, hold component stock and do the final kitting of each order. This lets the venture focus its resources on product design development, quality assurance and customer feedback prior to scaling operations. Grippy competes on Product Leadership (Section 2).

2. Competitive & Operational Strategy

2.1 Competitive Landscape and Positioning

The competitive landscape for professional tool-storage features is currently dominated by major players such as Stanley Black & Decker (including DEWALT) and Klein Tools, delivering standardized tool storage solutions. The barrier to entry is defined by these market incumbents, which leverage their established brand recognition, extensive distribution chains and consistent production through reliable manufacturing standards.

Stanley Black & Decker, the market leader, reported $13.3 billion of Tools & Outdoor sales in fiscal 2024[4]. Two canvas blocks explain its position: Channels, because home centers, mass merchants and distributors put its brands in front of every trade, and Key Resources, its global manufacturing scale and brands such as DEWALT. Klein Tools, a family-owned manufacturer founded in 1857, is the electrician specialist[5]; its position rests on Customer Segments and Value

[3]Felin, Gambardella and Zenger (2021), “Value Lab.”
[4]Stanley Black & Decker, Form 10-K, fiscal 2024.
[5]Klein Tools, company website, kleintools.com.

Proposition, trade-specific durability sold to electricians. The two also differ in cost structure and in how broad a range they serve (Exhibit 1). Each has a vulnerability: Stanley Black & Decker's scale model favors broad, standardized lines over niche designs, and Klein's bags extend a storage product rather than rethinking how electricians carry and reach their tools.

The closest direct competitor is Klein Tools' Tool Station Backpack, whose design features a built-in, fold-out station workplace layout. At about $110[6], it sets the price reference for Grippy. Grippy adds two features the Klein listing does not describe: the ergonomic support harness, which answers the back-support pain point raised in our interviews, and detachable compartments for access on the move.

2.2 Value Discipline and Operational Alignment

Grippy competes on Product Leadership, one of Treacy and Wiersema's three value disciplines[7]. Stanley Black & Decker competes on scale and cost and Klein on trade breadth; a startup making small outsourced batches cannot match either on cost, but it can lead on design. The order winners are functional design and ergonomics. The qualifiers are durability, a price near the Klein Tool Station and a reasonable lead time. The discipline drives the rest of the operating strategy: development is iterative (Section 3); the bag is modular, with outsourced sewing, in-house final kitting and Assemble-to-Order (Section 4); and quality is defined as functional performance (Section 5).

3. New Product Development Strategy

Grippy's development strategy has been mainly focused on introducing new and innovative solutions to the challenges electricians face when transporting, organizing and accessing their tools. Traditional backpacks often require electricians to choose between accessibility to their gear, capacity, ease of transport and comfort; Grippy aims to eliminate these trade-offs with three V1 features: a backpack that opens flat, detachable compartments and an ergonomic support harness. The strategy combines distinct value, shown through product and design, with building reliable small-batch operations for a successful first launch.

The first stage was to understand the needs of target customers. Our initial research found two linked problems: electricians struggle to reach their tools quickly while moving between sites,

[6]Keep Supply, Klein 55482 listing, accessed October 27, 2026.
[7]Treacy and Wiersema (1993), Harvard Business Review.

and the physical strain of carrying them grows as they get older. These findings led us to challenge the assumption that electricians must choose between storage on the go, quick and reliable access to their tools, and comfort.

The next step was a rough cardboard prototype, to visualize the product and start testing it with electricians. The cardboard prototype can already test tool retrieval time, tool visibility and how easily the bag opens flat; durability, load, carrying comfort and weight distribution need a sewn prototype. Test protocols and results are in Section 6. Comparing Grippy with traditional tool bags and backpacks shows whether the design improves on what is already on the market, which matters because Grippy's success depends on delivering the benefits its value proposition promises.

As the capabilities gap assessment in Section 6 shows, progressing from prototyping to commercialization will require capabilities Grippy does not yet have, especially in material sourcing, manufacturing, quality control and product distribution. Consistent with Grippy's Product Leadership discipline (Section 2), development is iterative: each prototype round targets the gaps the assessment identifies. To close them, Grippy will partner with reliable suppliers and manufacturers and develop its own standardized production process. Grippy borrows Stanley Black & Decker's operating practices (consistent manufacturing, an efficient supply chain) while competing on design.

Before committing to a full market launch, Grippy will run a Phase 0 pilot batch to evaluate manufacturing feasibility, product consistency and operational costs. This limited initial batch will identify potential bottlenecks and problems, confirm that quality control functions as expected and test distribution capabilities without over-committing resources. The results will determine whether the production system is reliable and efficient enough to support launch.

4. Operational Process & Supply Chain Strategy

4.1 Proposed Operating Model

Grippy’s proposed operating model shall focus on developing a durable, ergonomic and functional backpack for electricians, in line with the competitive priorities ranked in Section 2. As Grippy is in the early development stage, the initial production strategy shall focus on small-batch production, quality testing and flexibility as opposed to large-scale production.

The raw inputs (fabric, webbing, zippers, buckles and connectors) will be sourced from suppliers. A contract backpack manufacturer will cut and sew the bag bodies, straps and compartments to Grippy's specification; Grippy will inspect incoming lots, hold component stock and do the final kitting of each order.

4.2 Process Design

Proposed Production Workflow: Material sourcing → Manufacturer sews bodies and compartments → Inspection 1 (manufacturer) → Inspection 2 (Grippy goods-in) → Component stock [CODP] → Customer order → Grippy final kitting → Inspection 3 (opens flat, compartments attach, harness and strap load) → Ship

Grippy finalizes the open-flat design, compartments and harness and sources durable, light and readily available materials. The manufacturer cuts and sews bodies, straps and compartments in small lots and inspects them before shipping. Grippy checks each lot on arrival and holds it as component stock. When an order arrives, Grippy kits the chosen compartments with a body, checks that the bag opens flat, that the compartments attach securely and that the harness and straps carry their rated load, then packs and ships it.

Potential Inefficiencies. Major inefficiencies in the process of making the bags will likely include shortages of materials, delays with external suppliers, bad stitching/assembly and more time being taken to make the very specialized bags than normal.

Section 5.2 sets out the specifications, supplier coordination and in-process checks that reduce these risks.

4.3 Push vs. Pull Strategy

A push strategy produces in advance of demand to minimize customer waiting times, but for a new product it risks overproduction and unsold inventory.

A pull strategy starts production only once customers have ordered, which removes the risk of overproduction but can lengthen waiting times.

At launch Grippy runs Assemble-to-Order (Phase 1): bodies and compartments are made ahead of demand (push); final kitting starts only when an order arrives (pull).

As sales become more predictable, the company can adjust the quantities of products to be produced and hold a small inventory of the most popular configurations (Phase 2).

4.4 Customer Order Decoupling Point (CODP)

The Customer Order Decoupling Point (CODP) is the stage where production shifts from forecast-driven to customer-order-driven.

Three potential strategies for Grippy are Make-to-Stock (MTS), Assemble-to-Order (ATO) and Make-to-Order (MTO), compared in Exhibit 2.

Recommended Strategy: Assemble-to-Order (ATO). ATO is the most appropriate production strategy for Grippy because of the many modular parts: a base set of detachable compartments plus add-on compartments the customer picks at order, which create many compartment configurations.

Because there are so many compartment configurations that Grippy could offer, it would be impractical for the company to hold stock of complete products.

The CODP is located at the final kitting stage where available modules would be used to build the complete product based on the customer's order.

During the Phase 0 pilot batch, Grippy validates components and demand before moving to Assemble-to-Order.

4.5 Operational Coordination

Production Coordination. Grippy will set production schedules with the manufacturer and maintain standard specifications, so that materials and parts are available when required and quality is consistent from batch to batch.

Inventory Coordination. Inventory tracking covers bag bodies and compartments, keeping enough stock of frequently used items to avoid kitting delays while minimizing inventory costs.

Order and Fulfillment Coordination. Grippy will sell direct online to independent contractors first and through electrical supply houses later. Orders are managed through a single order management system that connects stock levels with final kitting and delivery. When an order is confirmed, the required components are reserved, kitted and dispatched as a complete product, with a target of shipping within three business days.

In summary, Grippy will use outsourced manufacturing with in-house final kitting and a formal production process, running Assemble-to-Order from launch and stocking the most popular

configurations once demand is stable (Phase 2). This keeps costs and inventory risk low while meeting the differing needs of electricians.

5. Capacity, Efficiency & Quality Management

5.1 Production Capacity and Resource Planning

At this stage of development, Grippy has a cardboard prototype that opens all the way until the bag is flat; it does not yet include the detachable compartments, which the next prototype adds. Grippy's own utility belt and a hanging point are V2 features. To become a sellable product, the bag needs durable materials such as heavy-duty nylon or polyester that can withstand the unpredictable nature of a job site, breathable mesh straps, and heavy-duty nylon stitching, because a bag that opens fully puts repeated stress on its seams.

Grippy will select a contract backpack manufacturer that can sew the open-flat construction and the support harness, accepts a low minimum order and has a record of consistent quality. Capacity then depends on two stages. Upstream, the manufacturer's minimum order and lead time set how much component stock Grippy must hold. Downstream, final kitting sets weekly output: the kitting hours the team can staff divided by the minutes each order takes. Because kitting one order is a short manual task, the likely bottleneck in Phases 0 and 1 is the manufacturer's replenishment lead time; kitting becomes the constraint only if orders outgrow the hours the team can staff. Until quotes arrive, Grippy plans on a 50-unit pilot, a lead time of up to eight weeks and about ten minutes of kitting per order, so ten staffed hours a week would kit about 60 orders.

5.2 Operational Efficiency and Capacity Improvements

Because the bag opens fully and has removable parts, Grippy will standardize product measurements, document its material and stitching specifications and fix the configuration of the removable parts. Setting guidelines for manufacturing reduces the likelihood of production errors, and it also creates a kitting process that can be easily repeated.

Performance will be tracked against six KPIs, each with a starting target that the pilot batch will refine: tool retrieval at least 30% faster than a conventional backpack, a 15 kg rated load, harness and straps tested to 1.5 times that load, a defect rate of 3% or less, final kitting within ten

minutes per order and shipping within three business days. Recording them for each prototype and for the pilot batch shows whether the process is improving.

5.3 Product Quality & Performance Measurement

Grippy's quality standard follows from its value proposition: each bag must carry its rated load, survive repeated open and close cycles and keep the harness securely attached under load. The cardboard prototype cannot show how the bag performs with repeated use, so these tests need sewn samples.

Grippy will prioritize feedback from electricians in the target market to find improvements or weaknesses and to check whether the product lives up to the standard promised to prospective customers. Their feedback will also show how to better implement back support, a pain point that has been recurring in our interviews thus far.

To gauge product quality and consistency, Grippy will inspect at three points that match the workflow in Section 4.2: the manufacturer's check before shipping, Grippy's goods-in check of each lot, and a final check of every kitted order. Recording the findings from each inspection will pinpoint recurring issues with suppliers or the production process and enable design or manufacturing changes where necessary.

6. Operational Risks & Implementation Strategy

6.1 Capability Gaps and Implementation Path

The audit compares what Product Leadership requires (Section 2.2) with what Grippy has today: a cardboard prototype, interview findings and no production, testing or sales capability (Exhibit 3). Design and ergonomics must beat the Klein Tool Station; durability, price and lead time only need to match it. Three gaps stand between the prototype and a sellable product:

- **Ergonomic design (build, with an adviser).** The harness is the main order winner, yet no team member is trained in ergonomics. Grippy keeps design in-house, as the source of differentiation, and has a physiotherapist check how the harness spreads a full tool load.
- **Manufacturing (partner).** Sewing an open-flat body and a load-bearing harness needs industrial machines that would sit idle at Grippy's volumes. A contract manufacturer turns that fixed cost into a unit cost, at the price of less control over quality and lead time.

- **Quality assurance (build, then buy).** Durability is a qualifier: a bag that fails early loses the customer whatever its design. Grippy builds simple load and open/close tests in-house and buys certified lab tests before launch.

Final kitting and online sales are smaller gaps that the team can learn during the pilot. Phase 0 (about three months) produces sewn samples, runs the tests in Section 6.2 and ends with a 50-unit pilot batch. Grippy moves to Phase 1, Assemble-to-Order sold direct online, only if the pilot's defect rate is 3% or lower and replenishment lead time is eight weeks or less. After three months of stable demand, Phase 2 stocks the most popular configurations and adds electrical supply houses. These figures are planning assumptions until quotes and test results replace them.

6.2 Deal-Killer Risks and Experiments

A deal-killer is an assumption that, if false, stops the plan whatever else goes right. Grippy has three, each with a kill threshold fixed before testing (Exhibit 4):

- **The access benefit is not real.** If opening flat does not make tools faster to reach, the value proposition fails. Test: the rapid prototype test below. Kill threshold: Grippy is less than 30% faster than a conventional backpack.
- **Unit cost is too high.** Product Leadership allows only a small premium over Klein's $110, so Grippy plans a $120 price. Test: quotes from three contract manufacturers. Kill threshold: a landed unit cost above $60 at 500 units, a gross margin below 50%, too thin to fund marketing, shipping and returns.
- **The bag cannot carry a full kit.** Opening flat puts a long seam and zipper under load. Test: load and open/close cycle tests on three sewn samples. Kill threshold: any failure below 1.5 times the 15 kg rated load, or within 1,000 cycles, about a year of daily use.

Rapid prototype test. Five participants each retrieve the same five tools from a conventional backpack and from the cardboard Grippy opened flat, alternating which bag comes first; each retrieval is timed and dropped tools are counted. Cardboard is enough because the test measures layout and visibility, not strength.

6.3 Value-Proposition Test

Hypothesis (Section 1.3): independent electricians who move between job sites will prefer, at a price close to the Klein Tool Station, a backpack that opens flat, carries detachable compartments for access on the move and reduces carrying strain through an ergonomic support harness.

Method: Grippy shows 15 independent electricians the prototype beside the Klein Tool Station's specifications, photos and $110 price, asks which they would buy if Grippy cost $120 and the most they would pay, and offers a refundable $20 pre-order deposit to separate polite interest from commitment. Kill threshold, set in advance: fewer than half prefer Grippy at $120, or fewer than three pay a deposit.

7. Strategic Recommendations & Conclusion

Grippy should compete on Product Leadership and shape its operations around that choice. First, launch one V1 backpack that opens flat, with detachable compartments and an ergonomic support harness, and hold the utility belt and hanging point for V2 so that development effort stays on the harness, the main order winner. Second, outsource sewing but keep design, quality testing and final kitting in-house, because that is where differentiation and the customer order decoupling point sit. Third, run Assemble-to-Order from launch and change phase only when the triggers in Section 6.1 are met. Fourth, sell direct online first, for margin and direct customer feedback, and add electrical supply houses once demand is stable.

The tests form a go/no-go sequence. Grippy orders the pilot batch only if the retrieval test, the manufacturer quotes and the load tests clear their kill thresholds, and launches only if the value-proposition test does. Each test costs little next to a production order, so the largest commitment comes last.

In short, electricians are forced to trade comfort and access for storage. Grippy believes they will pay to avoid that trade-off, competes on Product Leadership to deliver it, builds a modular Assemble-to-Order operation around the design, and tests each deal-killer before money is committed.

Exhibits

Exhibit 1. Business Model Canvas: Stanley Black & Decker and Klein Tools

Block	Stanley Black & Decker (incl. DEWALT)	Klein Tools
Customer Segments	All trades and DIY consumers	Professional electricians and allied trades
Value Proposition	Broad, reliable tool and storage range at competitive prices	Trade-specific durability and electrician-focused design
Channels	Home centers, mass merchants, distributors, online retail	Electrical distributors, home centers, online retail
Customer Relationships	Brand loyalty built through tool platforms	Long-standing trust among electricians
Revenue Streams	Power tools, hand tools, storage and outdoor products	Hand tools, testers, tool bags and storage
Key Resources	Global manufacturing scale; brands such as DEWALT, STANLEY and CRAFTSMAN	Brand reputation with electricians; specialist product design
Key Activities	High-volume manufacturing, supply-chain management, marketing	Product design and quality manufacturing for the electrical trade
Key Partners	Large retailers, suppliers, contract manufacturers	Electrical distributors and suppliers
Cost Structure	Scale-driven, low unit cost	Higher unit cost, premium positioning

Decisive blocks: Channels and Key Resources for Stanley Black & Decker; Customer Segments and Value Proposition for Klein Tools. Sources: footnotes 4 to 6.

Exhibit 2. Customer order decoupling point options for Grippy

Strategy	Application to Grippy	Advantages	Limitations
Make-to-Stock (MTS)	Finished Grippy backpacks are manufactured and stored before orders arrive.	Fast delivery and efficient batch production.	Higher inventory risk and limited flexibility.
Assemble-to-Order (ATO)	Standard bag components are prepared in advance and assembled after an order is received.	Balances customization, inventory control, and delivery speed.	Requires component availability and efficient assembly coordination.

Make-to-Order (MTO)	Manufacturing begins only after a customer places an order.	Low finished-goods inventory and greater customization.	Longer delivery times and potentially higher unit costs.

Exhibit 3. Strategic operational audit and capability gaps

Requirement (Section 2.2)	**Current capability**	**Gap**	**Decision and reason**
Ergonomic harness that beats Klein (order winner)	Interview findings; cardboard prototype	Large	Build, with a physiotherapist adviser: the harness is the differentiator
Open-flat body sewn to specification (order winner)	None	Large	Partner with a contract manufacturer: machines too costly at low volume
Durability equal to Klein (qualifier)	No standards or test equipment	Large	Build simple tests; buy certified lab tests before launch
Price near Klein's $110 (qualifier)	No cost data	Medium	Collect three manufacturer quotes; plan a $120 price
Reasonable lead time (qualifier)	No kitting or fulfillment process	Small	Build in-house: kitting is simple and holds the CODP
Direct online sales	None	Small	Buy a standard e-commerce platform

Exhibit 4. Experiment protocols and kill thresholds

Deal-killer	**Test**	**Prototype**	**Sample**	**Kill threshold**
Access benefit not real	Timed tool retrieval against a conventional backpack	Cardboard (now)	5 people, 5 tools each	Grippy less than 30% faster
Unit cost too high	Manufacturer quotes	Technical drawings	3 manufacturers	Landed cost above $60 at 500 units
Bag cannot carry a full kit	Load and open/close cycle tests	Sewn samples	3 samples	Failure below 22.5 kg (1.5 x 15 kg) or within 1,000 cycles
Value proposition fails	Side-by-side with Klein; refundable pre-order deposit	Cardboard and photos	15 electricians	Under half prefer Grippy at $120, or fewer than 3 deposits

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